

## YERMAKOVSKOYE, Russian Federation

WMO: 298690

Lat: 53.270N

Lon: 92.403E

Elev: 977

StdP: 14.18

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -34.0      | -28.1 | -37.7                         | 0.7 | -33.8 | -31.9 | 0.9 | -28.0 | 19.2                  | 18.8 | 15.8 | 17.1 | 0.0                   | 50   | 0.549 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 19.3                   | 85.9            | 67.6 | 82.9 | 66.3 | 79.8 | 64.8 | 70.1                | 82.0 | 68.4 | 79.4 | 66.7 | 76.5 | 3.6                  | 50   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 66.1                            | 99.7 | 75.1 | 64.5 | 94.1 | 73.7 | 62.9 | 89.0 | 72.0 | 34.8          | 82.4 | 33.3 | 79.6 | 32.0 | 76.5 | 78.8              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |     |                    |      | n-Year Return Period Values of Extreme Temperature |     |            |      |            |      |            |      |       |      |
|-------------------|------|------|----------------------------|-----|--------------------|------|----------------------------------------------------|-----|------------|------|------------|------|------------|------|-------|------|
|                   |      |      | Mean                       |     | Standard Deviation |      | n=5 years                                          |     | n=10 years |      | n=20 years |      | n=50 years |      |       |      |
| 1%                | 2.5% | 5%   | Min                        | Max | Min                | Max  | Min                                                | Max | Min        | Max  | Min        | Max  | Min        | Max  |       |      |
| (a)               | (b)  | (c)  | (e)                        | (f) | (g)                | (h)  | (i)                                                | (j) | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |       |      |
| (4)               | 15.6 | 12.7 | 10.2                       | DB  | -37.7              | 91.5 | 6.6                                                | 2.3 | -42.4      | 93.1 | -46.2      | 94.4 | -49.9      | 95.7 | -54.7 | 97.4 |
| (5)               |      |      |                            | WB  | -36.8              | 74.2 | 6.6                                                | 2.4 | -41.6      | 75.9 | -45.5      | 77.3 | -49.2      | 78.6 | -54.1 | 80.3 |

## Monthly Climatic Design Conditions

|      |                                                                   | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |       |      |
|------|-------------------------------------------------------------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|      |                                                                   | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |       |      |
| (6)  | Temperatures, Degree-Days and Degree-Hours                        | DBAvg       | 36.3  | 0.0   | 7.1   | 25.2  | 40.6  | 52.0  | 63.6  | 66.9  | 62.5  | 50.6  | 37.2  | 21.3  | 6.6   | (6)  |
| (7)  |                                                                   | DBStd       | 25.11 | 15.25 | 14.57 | 11.80 | 8.72  | 8.22  | 6.69  | 5.02  | 6.16  | 7.31  | 8.30  | 12.56 | 14.76 | (7)  |
| (8)  |                                                                   | HDD50       | 6589  | 1551  | 1202  | 770   | 297   | 78    | 2     | 0     | 1     | 80    | 401   | 861   | 1346  | (8)  |
| (9)  |                                                                   | HDD65       | 10694 | 2016  | 1622  | 1235  | 731   | 410   | 103   | 38    | 122   | 435   | 860   | 1311  | 1811  | (9)  |
| (10) |                                                                   | CDD50       | 1580  | 0     | 0     | 0     | 16    | 139   | 410   | 524   | 388   | 97    | 6     | 0     | 0     | (10) |
| (11) |                                                                   | CDD65       | 209   | 0     | 0     | 0     | 0     | 6     | 61    | 96    | 44    | 2     | 0     | 0     | 0     | (11) |
| (12) |                                                                   | CDH74       | 2372  | 0     | 0     | 0     | 12    | 179   | 748   | 865   | 509   | 58    | 1     | 0     | 0     | (12) |
| (13) |                                                                   | CDH80       | 608   | 0     | 0     | 0     | 2     | 37    | 207   | 220   | 130   | 12    | 0     | 0     | 0     | (13) |
| (14) | Wind                                                              | WSAvg       | 2.8   | 1.7   | 2.3   | 3.5   | 3.8   | 3.8   | 3.1   | 2.5   | 2.3   | 2.6   | 2.7   | 3.0   | 2.5   | (14) |
| (15) | Precipitation                                                     | PrecAvg     | 21.4  | 0.7   | 0.6   | 0.7   | 1.4   | 1.9   | 3.0   | 3.2   | 3.1   | 2.6   | 1.8   | 1.4   | 1.0   | (15) |
| (16) |                                                                   | PrecMax     | 31.7  | 1.5   | 1.3   | 1.9   | 2.6   | 3.9   | 5.3   | 6.9   | 7.2   | 9.2   | 5.1   | 3.5   | 2.4   | (16) |
| (17) |                                                                   | PrecMin     | 15.6  | 0.2   | 0.1   | 0.2   | 0.2   | 0.2   | 1.1   | 1.7   | 1.1   | 0.6   | 0.4   | 0.4   | 0.5   | (17) |
| (18) |                                                                   | PrecStd     | 3.7   | 0.4   | 0.3   | 0.4   | 0.6   | 0.9   | 1.1   | 1.1   | 1.4   | 1.6   | 0.9   | 0.7   | 0.6   | (18) |
| (19) | Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB    | 34.6  | 39.2  | 56.6  | 74.7  | 84.6  | 89.5  | 88.8  | 88.3  | 80.8  | 66.8  | 50.0  | 36.4  | (19) |
| (20) |                                                                   | 0.4%        | MCWB  | 30.5  | 32.8  | 42.8  | 53.1  | 60.4  | 66.8  | 70.9  | 70.9  | 64.3  | 52.3  | 41.3  | 31.4  | (20) |
| (21) |                                                                   | 2%          | DB    | 30.2  | 35.0  | 50.7  | 67.9  | 79.4  | 85.3  | 85.4  | 83.7  | 73.4  | 59.2  | 44.5  | 32.6  | (21) |
| (22) |                                                                   | 2%          | MCWB  | 26.7  | 30.0  | 40.4  | 50.2  | 57.5  | 66.1  | 69.2  | 68.4  | 59.9  | 49.0  | 38.6  | 28.5  | (22) |
| (23) |                                                                   | 5%          | DB    | 26.4  | 31.7  | 45.9  | 62.4  | 74.7  | 81.9  | 82.4  | 79.7  | 68.6  | 53.8  | 40.5  | 30.2  | (23) |
| (24) |                                                                   | 5%          | MCWB  | 23.6  | 27.8  | 37.5  | 48.0  | 55.5  | 64.9  | 67.9  | 66.1  | 57.5  | 46.0  | 35.8  | 26.9  | (24) |
| (25) |                                                                   | 10%         | DB    | 22.0  | 27.7  | 41.3  | 57.0  | 69.8  | 78.3  | 79.0  | 76.0  | 64.1  | 50.0  | 37.0  | 26.5  | (25) |
| (26) |                                                                   | 10%         | MCWB  | 19.6  | 24.1  | 34.6  | 45.4  | 54.0  | 63.5  | 66.3  | 64.2  | 55.2  | 44.1  | 33.3  | 24.1  | (26) |
| (27) | Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB    | 31.8  | 34.0  | 44.6  | 54.5  | 63.3  | 70.7  | 73.1  | 72.1  | 65.3  | 53.7  | 42.9  | 32.2  | (27) |
| (28) |                                                                   | 0.4%        | MCDWB | 33.6  | 37.7  | 54.8  | 71.3  | 80.3  | 84.1  | 85.3  | 85.3  | 75.9  | 63.4  | 48.1  | 34.9  | (28) |
| (29) |                                                                   | 2%          | WB    | 27.1  | 30.6  | 40.8  | 51.5  | 60.1  | 68.3  | 70.7  | 69.2  | 61.6  | 50.0  | 39.2  | 29.1  | (29) |
| (30) |                                                                   | 2%          | MCDWB | 30.0  | 34.7  | 49.9  | 64.5  | 74.6  | 81.5  | 82.2  | 81.2  | 71.4  | 57.2  | 43.8  | 32.2  | (30) |
| (31) |                                                                   | 5%          | WB    | 23.6  | 27.4  | 37.8  | 48.7  | 57.5  | 66.4  | 69.1  | 67.2  | 58.9  | 47.0  | 36.3  | 26.8  | (31) |
| (32) |                                                                   | 5%          | MCDWB | 26.2  | 31.7  | 45.4  | 60.3  | 70.7  | 78.2  | 79.6  | 77.7  | 66.7  | 53.3  | 40.2  | 29.9  | (32) |
| (33) |                                                                   | 10%         | WB    | 19.7  | 24.2  | 35.1  | 45.9  | 54.9  | 64.6  | 67.5  | 65.1  | 56.1  | 44.2  | 33.3  | 24.0  | (33) |
| (34) |                                                                   | 10%         | MCDWB | 22.1  | 27.7  | 41.0  | 56.1  | 66.5  | 75.3  | 76.8  | 73.9  | 62.4  | 49.3  | 36.6  | 26.5  | (34) |
| (35) | Mean Daily Temperature Range                                      |             | MDBR  | 19.0  | 22.0  | 22.1  | 21.6  | 22.7  | 21.8  | 19.3  | 20.1  | 20.5  | 17.5  | 14.1  | 16.1  | (35) |
| (36) |                                                                   | 5% DB       | MCDBR | 15.9  | 21.2  | 27.2  | 34.1  | 34.0  | 29.5  | 25.6  | 27.8  | 30.4  | 26.9  | 16.9  | 15.3  | (36) |
| (37) |                                                                   |             | MCWBR | 13.1  | 16.5  | 17.3  | 17.8  | 14.7  | 12.4  | 11.1  | 13.4  | 16.7  | 16.1  | 11.5  | 12.3  | (37) |
| (38) |                                                                   | 5% WB       | MCDWB | 15.2  | 20.5  | 25.1  | 30.6  | 27.6  | 24.4  | 22.4  | 25.1  | 27.5  | 25.5  | 15.7  | 14.7  | (38) |
| (39) | MCWBR                                                             |             | 12.8  | 16.1  | 16.3  | 16.9  | 13.0  | 10.9  | 10.3  | 12.6  | 15.8  | 16.0  | 11.1  | 12.1  | (39)  |      |
| (40) | Clear-Sky Solar Irradiance                                        | taub        | 0.284 | 0.320 | 0.367 | 0.411 | 0.418 | 0.403 | 0.413 | 0.399 | 0.363 | 0.350 | 0.325 | 0.290 | (40)  |      |
| (41) |                                                                   | taud        | 2.056 | 2.012 | 1.985 | 2.122 | 2.178 | 2.329 | 2.326 | 2.338 | 2.375 | 2.277 | 2.102 | 2.005 | (41)  |      |
| (42) |                                                                   | Ebn at Noon | 207   | 237   | 251   | 258   | 263   | 269   | 264   | 260   | 256   | 229   | 188   | 174   | (42)  |      |
| (43) |                                                                   | Edh at Noon | 23    | 34    | 44    | 44    | 43    | 38    | 37    | 35    | 30    | 26    | 23    | 21    | (43)  |      |
| (44) | All-Sky Solar Radiation                                           | RadAvg      | 341   | 670   | 1110  | 1407  | 1650  | 1921  | 1747  | 1457  | 1030  | 584   | 298   | 224   | (44)  |      |
| (45) |                                                                   | RadStd      | 27    | 45    | 73    | 86    | 146   | 189   | 130   | 117   | 104   | 53    | 37    | 23    | (45)  |      |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |       |       |       |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|-------|-------|-------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
| (d)  | (e)                     | (f)   | (g)     | (h)    | (i)     | (j)   | (k)   | (l)         | (m)   |       |       |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) | Regional (12 neighbors) | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page